TARUN MODEKURTY

Machine Learning Engineer | Industrial AI & Computer Vision
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SUMMARY

Software engineering co-op student who takes ML from raw data to the factory floor: built a weld-quality early-warning system on **3.75M production welds** (and earned welding certification to understand them), a video model that **beats the published state of the art** at recognizing actions it has never seen, and the backend of an AI-compliance risk platform.

EDUCATION

Bachelor of Engineering, Software Engineering (Co-op) | *Ontario Tech University, Oshawa, ON* Expected May 2028

EXPERIENCE

Application Engineering Co-op (ML & Industrial Data) | *Multimatic Dynamic Suspensions* May 2026 – Present

- Hired for SCADA data analysis; expanded the scope into a weld-quality early-warning system covering **3.75 M production welds** from 8 FANUC welding robots, being evaluated by operators to prioritize inspections each shift and catch weld defects
- Found that high frequency waveform data shows weld defects, before visual inspection. Achieved a **70 percent** catch rate with minimal **false alarms** across **3.75M** production welds, so operators can trust every alert
- Built a daily "inspect these 30 first" list from an ML defect-risk model (**AUC 0.81**) trained on 984 labels I hand-corrected with weld process engineers, so inspectors check the riskiest parts before they ship
- Built the real-time telemetry pipeline from scratch on an air-gapped factory network (MQTT), streaming live current and voltage waveforms from all 8 welders to a dashboard the team uses daily

Research Assistant, Computer Vision | *Visual Computing Lab, Ontario Tech University* Feb 2026 – Present

- Built a video model that recognizes actions it has never seen (e.g., "folding paper"), known as Zero-Shot Compositional Action Recognition. It pairs a frozen video expert for verbs with a frozen image expert for objects and fine-tunes no backbone. It scores **3.1 points above** RCORE, the published state of the art on Sth-com, at matched backbones; paper in preparation for CVPR
- Ran controlled ablations proving the gain comes from the architecture rather than the backbone, including swapping in RCORE's own CLIP backbone. They showed the video expert nearly doubles verb accuracy on unseen actions **(64% vs. 34%)**
- Trained and evaluated on **79K** videos across 3 seeds using a national A100 GPU cluster (Narval, SLURM)

Research Associate | *AI Hub, Durham College* Feb 2026 – May 2026

- Benchmarked 3 embedding models on 630 manufacturing non-conformance and complaint records and recommended **all-MiniLM-L6-v2 (0.1720 vs. 0.1332 MRR; 29% higher)** for similar-defect search

AI Software Engineer | *METRIS by Sanjeevani AI, "the FICO score for AI compliance"* Dec 2025 – Mar 2026

- Led the backend (FastAPI) for an AI-governance platform that scores a company's regulatory risk against **2,100+ checkpoints** (EU AI Act, GDPR, ISO 42001, NIST); completing a full assessment in under **40 minutes** vs. weeks for a manual audit
- Designed the 35-agent engine that turns compliance gaps into a dollar-exposure estimate ($M) so risk leaders can prioritize fixes by $ impact; owned orchestration design

Generative AI Engineering Intern | *Second Brain Labs* May 2025 – Oct 2025

- Built a RAG tool that mined **75,000+ sales conversations**, separating calls that closed from calls that didn't, to surface **1,800+ objection-handling patterns** that won deals, so the sales team could deploy a chatbot that tailored each reply to the prospect's objection, like offering deals on price pushback and comparison guides on competitor mentions

PROJECTS

GridSens: 1st Place of 15 teams, Cognitive Innovation Competition | *GNN, React, FastAPI* Apr 2026

- Built a graph neural network that warns of cascading power-grid failures 2–4 time steps before they happen; demonstrated by replaying the 2021 Texas winter storm in simulation on real ERCOT load data (peak 73,650 MW)

CodeGuard: Multi-Agent Code Review | *DeepSeek Coder, CodeLlama, Qwen* Dec 2025

- Built a code reviewer where 3 LLMs must agree before a finding is raised, cutting false positives; raised detection accuracy from **1 of 7** (keyword search) to **7 of 7** (semantic search) seeded security bugs.

SKILLS

Machine Learning: PyTorch, scikit-learn, time-series anomaly detection, computer vision, RAG and LLM agents (LangGraph)
Data & Systems: Python, SQL, FastAPI, MQTT, Docker, AWS, SLURM/HPC
Domain: Industrial telemetry & SCADA (Ignition), statistical process control, AI regulatory frameworks (EU AI Act, ISO 42001, NIST)
Certifications: Manual and robotic welding (Multimatic, 2026)